

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

- A. It activates TLR5 on the surface of macrophages.
- B. It primarily induces the Th1-cell response.
- C. Peptidoglycan is its major antigen that induces an immune response.
- D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think> To answer this question , we need to look at the symptoms and diagnosis of the patient .

The patient is diagnosed with community-acquired pneumonia , which is a type of lung infection that is typically caused by bacteria .

The patient's symptoms , such as sore throat and cough , are consistent with a bacterial infection , and the presence of patchy reticular opacities in the perihilar regions of both lungs on the X-ray is also consistent with a bacterial infection .

The likely cause of the patient's condition is a bacterial pathogen , such as Streptococcus pneumoniae , which is a common cause of community-acquired pneumonia .

The immune response elicited by a bacterial pathogen such as this is likely to involve the activation of the innate immune response , which includes the activation of TLR5 on the surface of macrophages , leading to the production of pro-inflammatory cytokines and the activation of the adaptive immune response .

Therefore , based on the likely cause of the patient's condition , it is most likely that the immune response elicited by the pathogen is the activation of TLR5 on the surface of macrophages .</think><answer> It activates TLR5 on the surface of macrophages .